

**U.S. INTERNATIONAL TRADE COMMISSION
WASHINGTON, D.C.**

**Before the Honorable Doris Johnson Hines
Administrative Law Judge**

In The Matter Of

**CERTAIN SMART WEARABLE
DEVICES**

Investigation No. 337-TA-1478

**RESPONDENTS OURARING INC. AND OURA HEALTH OY'S
OPENING CLAIM CONSTRUCTION BRIEF**

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Respondents Ouraring, Inc. and Oura Health Oy (collectively, “Oura”) submits this Opening Claim Construction Brief regarding the construction of claim terms in the Asserted Patents.¹

I. INTRODUCTION

This investigation relates to four Samsung patents, but only two are at issue for purposes of claim construction—the ’675 and ’359 Patents. And while the parties dispute three total terms, the first and second terms rise and fall together. The final term “biometric signal” asks the Court to cure an error that both parties agree exists but dispute how to cure it.

To begin, the disputed ’675 Patent terms suffer from two independent and fatal flaws. First, neither the claim language, specification, nor prosecution history provides any objective boundaries for the claimed “first period of time,” “second period of time,” or “third period of time,” leaving a person of skill in the art without any means of determining what these time periods encompass. Second, the terms of degree “shortly before and/or immediately after” further compound the issue by requiring a temporal comparison between two periods of time that are themselves undefined. The specification offers no standard by which to measure “shortly” or “immediately,” rendering these terms dependent on the subjective nature of any one person’s opinion. Thus, as explained in greater detail below, the ALJ should adopt Oura’s proposed construction and find these terms to be indefinite.

Next, the constructions proposed by both parties with regards to the ’375 Patent seek to cure an undisputed error—the lack of antecedent basis for the term “the biometric signal.” Samsung seeks to not only fix the clear error but also amend the claim language (*i.e.*, change “the

¹ U.S. Patent Nos. 10,642,359 (the “’359 patent”), 10,231,675 (the “’675 patent”), 10,945,677 (the “’677 patent”), and 10,978,789 (the “’789 patent”) (collectively, “the Asserted Patents”).

biometric signal” to be “a biological signal”). No basis exists for this proposed amendment. Thus, as explained in greater detail below, the ALJ should reject Samsung’s proposed amendment and adopt the amendment offered by Oura (*i.e.*, change “the biometric signal” to be “a biometric signal”).

II. BACKGROUND

A. The Asserted Patents For Which A Claim Construction Dispute Exist

1. The '675 Patent

Samsung filed the foreign priority application that led to the '675 Patent on August 12, 2015.² The patent bears the title Method For Detecting Biometric Information and Electronic Device Using the Same. The USPTO issued the '675 Patent on March 19, 2019.

The title of this patent suggests that the disclosure concerns electronic devices capable of detecting biometric information. While the specification provides a laundry list of electronic devices, the claims focus on wearable electronic devices. '675 Patent at 46:31-48:24. For those types of devices, the specification includes examples such as a head-mount device (HMD), electronic glasses, electronic clothing, an electronic bracelet, an electronic necklace, an electronic appcessory, an electronic tattoo, a smart mirror, and a smart watch. *Id.* at 5:27-32. It also explains that the wearable device may include a bracelet, a strip, a band, an attachment type (band-aid type) band, a belt, ear-mounted earphones, headphones, a clothing type electronic device, a shoe type electronic device, an HMD, a hat type electronic device, a glove type electronic device, a finger-tip type electronic device, a clip type electronic device, an arm band type electronic device, a contact lens device, digital clothing, and a remote controller. *Id.* at 14:3-15.

² The '677 Patent shares a priority claim and specification with the '675 Patent. While this discussion is relevant to the '677 Patent, Oura provides citation to the '675 patent because the parties dispute the terms in the claims of the '675 Patent.

The patent explains that the electronic devices can include sensors that detect motion of the device and information about a user. *Id.* at 2:4-11. The types of user information that the device may detect can include blood pressure, heart rate, an electrocardiogram, skin moisture, saturation of peripheral oxygen, and the like. *Id.* at 1:33-35. The specification explains that the electronic devices may have difficulty measuring biometric information for a long period of time, so it states that it is describing methods for selectively operating a biometric sensor based upon a selected or specified condition. *Id.* at 1:59-63. The specification then seeks to describe various interrelationships between the operation of the motion sensor and biological sensor during different time periods. *Id.* at 2:4-36. As described in detail below, some of the claim language used in the '675 Patent does not allow for an understanding of several aspects as to how the method operates as set forth in the bolded claim language below. An example of the issue is the method that the '675 Patent seeks to describe in claim 13:

13. A method comprising:

determining whether an electronic device is worn on a part of a user's body using at least one first sensor of the electronic device;

acquiring first information on a motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user's body;

activating at least one second sensor based on at least a part of the first information on the motion of the electronic device;

detecting a biometric information of the user using the activated at least one second sensor;

during detecting the biometric information, acquiring second information on a motion of the electronic device using at least the part of the at least one first sensor; and

in response to identifying that the motion of the electronic device included in the second information is more than a threshold motion, inactivating the at least one second sensor.

Id. at 46:31-52 (emphasis added).

2. *The '359 Patent*

Samsung Ltd. filed the priority application that led to the '359 Patent on August 20, 2013—before companies like Oura introduced the present class of wearable ring devices that provide motion and biological information about a user. The patent bears the title “Wearable Biosignal Interface and Method Thereof” and issued on May 5, 2020, well after Oura had established the market for wearable electronic devices in the form of a ring. The specification describes a wearable biosignal interface in forms consistent with the products that existed in 2013, *e.g.*, watch or bracelet. *See* '359 Patent at Figure 1, 4:26-51, 5:5-7. The wearable biosignal interface may include a motion sensor that may detect motion of a body part, such as an arm. *Id.* at 4:45-5:7. It also may include a biosignal sensor. *Id.* at 5:5-7. The biosignal sensor may obtain a biosignal signal “associated with biological information such as, for example, muscle contraction and muscle relaxation occurring in the target.” *Id.* at 5:14-19, 10:54-59. The muscles targeted for measurement may include striated, cardiac, and smooth muscles. *Id.* at 3:65-4:3. While the patent does mention the potential to obtain information concerning the contraction and relaxation of the heart, it is devoid of any disclosure of other types of measured information, *e.g.*, blood oxygen level, resting heart rate. *Id.* at 3:55-59, 5:44-47 (discussing using an electromyogram (EMG) sensor to detect heart muscle contraction and relaxation).³

The patent describes using the information gathered by the motion sensor and the biosignal sensor to determine the validity of the signals and in cases where the signals are determined to be valid, to undertake an activity that controls an external device. *Id.* at 5:66-6:6 (describing use of the biosignal to determine whether the motion signal appeared intentional or unintentional to

³ An EMG sensor is a test that measures the electrical activity of a muscle. Rosing Dec. at ¶ 35.

determine the intention of a user to operate an external device). The external devices described in the specification include a mobile device, a television screen, a DVD player, a radio, a heating thermostat, a vehicle door, and the like. *Id.* at Figure 3, 6:64-7:2, 9:61-10:3. Some of these devices are the type that typically seek to verify the user that may operate the device, *e.g.*, mobile phones, heating thermostats, and entry to vehicles. Rosing Dec. at ¶ 37.

The '359 Patent claims the concept of an electronic device using information from a motion sensor and a biological sensor to perform a specified function, *e.g.*, control an external device. More specifically, claim 14 reads as follows:

14. A wearable device comprising:

a motion sensor;

a biological sensor; and

a controller configured to:

obtain a movement signal of an object external to the wearable device using the motion sensor;

obtain **a biological signal** of the object using the biological sensor;

determine a select period, of the movement signal, that meets a specified condition;

determine, as a valid signal, at least a portion of **the biometric signal** that is determined to correspond to the determined select period; and

perform a specified function using the valid signal.

'359 Patent at 15:14-27 (emphasis added).

III. LEGAL STANDARDS

Courts construe claims according to the meaning they would have to a person of ordinary skill in the art (“POSITA”) as of the filing date of the earliest application, in view of the intrinsic evidence: the claims, the specification, and prosecution history. *See Phillips v. AWH Corp.*, 415 F.3d 1303, 1312-13 (Fed. Cir. 2005). As a general rule, “terms in the claim are to be given their

ordinary and accustomed meaning.” *K-2 Corp. v. Salomon S.A.*, 191 F.3d 1356, 1362–63 (Fed. Cir. 1999) (internal citations omitted) (“[T]he ordinary and accustomed meaning of a disputed claim term is presumed to be the correct one....”). Although claim construction is ultimately a question of law, it is based on underlying factual considerations. *Teva Pharms. USA, Inc. v. Sandoz, Inc.*, 574 U.S. 318, 326-27 (2015). The specification “is always highly relevant to the claim construction analysis.” *Vitronics Corp. v. Conceptronic, Inc.*, 90 F.3d 1576, 1582 (Fed. Cir. 1996). “The construction that stays true to the claim language and most naturally aligns with the patent’s description of the invention will be, in the end, the correct construction.” *Phillips*, 415 F.3d at 1316 (citation omitted). Courts may depart from the plain and ordinary meaning only when a patentee acts as his own lexicographer or when the patentee disavows the full scope of a claim term either in the specification or during prosecution. *Thorner v. Sony Comput. Ent. Am. LLC*, 669 F.3d 1362, 1365 (Fed. Cir. 2012).

After reviewing the intrinsic evidence, courts may consider extrinsic evidence, including expert testimony, dictionaries, and treatises. *See Phillips* at 1314. Extrinsic evidence consists of all evidence external to the patent and prosecution history, including dictionaries and learned treatises. *Vitronics*, 90 F.3d at 1584. Extrinsic evidence is “less significant than the intrinsic record” because it is generally “less reliable.” *Phillips*, 415 F.3d at 1317-18. Courts should disregard extrinsic evidence that is at odds with the intrinsic record. *Id.* at 1317.

While the courts have established the above rules for construing disputed claim terms, they do not always lead to accomplishing the goal of providing a construction that meets the definiteness standard stated within our patent statute. A patent’s specification must “conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor regards as the invention.” 35 U.S.C. § 112(b). A claim is indefinite if, when “read in light

of the specification” and “prosecution history,” it “fail[s] to inform, with reasonable certainty, those skilled in the art about the scope of the invention.” *Nautilus, Inc. v. Biosig Instruments, Inc.*, 572 U.S. 898, 901 (2014). Because language has “inherent limitations,” the “reasonable certainty” standard exists to strike a “delicate balance,” “afford[ing] clear notice of what is claimed,” while recognizing those inherent limitations. *Id.* at 909 (quoting *Festo Corp. v. Shoketsu Kinzoku Kogyo Kabushiki Co.*, 535 U.S. 722, 731 (2002)). The definiteness requirement thus “mandates clarity, while recognizing that absolute precision is unattainable.” *Id.* at 910.

IV. PERSON OF ORDINARY SKILL IN THE ART

For the '675 and '359 Patents, a POSITA would have had at least a four-year degree in electrical engineering, mechanical engineering, biomedical engineering, computer engineering, or a related field of study, or equivalent experience, and at least two years of experience in academia or industry studying or developing wearable monitoring devices such as non-invasive biosensors. Additional work experience can make up for a lack of education, and additional education can make up for a lack of work experience. A POSITA could have also worked as part of a multidisciplinary team and would have drawn upon not only his or her skills, but also taken advantage of specialized skills of others on the team.

V. ANALYSIS OF DISPUTED CLAIM TERMS

A. Term No. 1: “acquiring first information on a motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user's body”

Claim Term	Complainant’s Construction	Respondents’ Construction
“acquiring first information on a motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user's body” Claim 13 of the '675 Patent	Within this phrase, “second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user’s body” mean: “second period of time which begins: (i) before the first period of time has ended when it is determined that the electronic device is worn, or (ii) immediately after the first period of time has ended when it is determined that the electronic device is worn”	Indefinite

This claim term presents two impediments to the task described in *Nautilus, Inc.* and its progeny—“provid[ing] objective boundaries for those of skill in the art.” *Interval Licensing LLC v. AOL, Inc.*, 766 F.3d 1364, 1371 (Fed. Cir. 2014) (citing *Nautilus, Inc.*, 572 U.S. at 911). First, neither the claim language nor the specification allows the skilled artisan to define the “first period of time” and the “second period of time” for which the temporal comparison of “shortly before” or “immediately after” must be assessed. Second, the use of terms of degree further exacerbates the tasks of providing “reasonable certainty” to the term’s meaning. At bottom, nothing in the intrinsic evidence suggests the meaning of the term “acquiring first information on a motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user's body,” rendering claim 13 indefinite. “The claims,

when read in light of the specification and the prosecution history, must provide objective boundaries for those of skill in the art.” *Interval Licensing*, 766 F.3d at 1371 (citing *Nautilus, Inc. v. Biosig Instruments, Inc.*, 572 U.S. 898, 911 (2014)).

Complainant seeks to address the concerns of this claim language through two tactics. First, it turns a blind eye to a key issue—defining the “first period of time” and the “second period of time” required by the claim language—by reciting those terms without providing a manner to understand what they encompass. Second, it rewrites the claim to remove subjective terminology, *i.e.*, eliminating the “shortly” from “shortly before.” Thus, the disputed term is indefinite for two independent reasons.

1. *The “First Time Period” and “Second Time Period” Terms are Indefinite*

Regarding the indefinite nature of the “first time period” and “second time period” terms, Complainant’s infringement contentions in the matter provide a concrete example of the problem with the claim language. Respondent offers this information to further illustrate the problem with the intrinsic record’s failure to provide support that allows a person of skill in the art to understand the “first period of time” and “second period of time” required by the claim.⁴

To begin, Complainant asserts that:

The Oura Accused Product includes the first sensor, such as an accelerometer, for acquiring first information on a motion of the electronic ring. The electronic ring and its accelerometer acquires this information for a second period of time, such as the period of time shortly before or immediately after it is determined that the electronic device is worn on a party of the user’s body, **where the first period of time is a period of time when the electronic ring is not being worn.**

⁴ Respondents recognize that claim construction proceedings are not the appropriate forum for resolving questions of noninfringement and do not raise such arguments here. That said, Respondents provide this tribunal with insight into Complainant’s position so that it may appreciate the importance of the dispute and as evidence of the absence of the required “reasonable certainty” for these claim terms that allows Complainant to offer any definition of the first, second, and third time periods that it deems necessary to assert an infringement allegation.

Ex. 2 at 8 (emphasis added). Here, Complainant alleges that the first period of time is *any* period of time when the electronic ring is not being worn. Complainant goes on to allege that the “second period of time” encompasses the time when a user is wearing the ring and may end when the user wakes up. *See* Ex. 2 at 36 (“The Oura Accused Product, during a third period of time beginning immediately after the second period of time (e.g., the period of time after the user wakes up)”). These allegations are purely subjective and unsupported by the specification. But more than that, they demonstrate that the claimed “first period of time” and “second period of time” are “facially subjective.” *See Interval Licensing*, 766 F.3d at 1373. The language of this limitation frustrates the important policy goal of “providing clarity such that a person of ordinary skill in the art [can] determine whether or not an accused product or method infringes the claim.” *Niazi Licensing Corp. v. St. Jude Med. S.C., Inc.*, 30 F.4th 1339, 1346-47 (Fed. Cir. 2022).

For example, the ’675 Patent explains:

According to an embodiment, the memory may store instructions that configure the processor to: determine whether the electronic device is worn on (or close to) a part of the user’s body using the at least one first sensor; **acquire information on the motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time selected (or specified) passes from the time when the electronic device is worn on (or close to) a part of the user’s body**; and activate (operate or trigger) the at least one second sensor based on at least a part of the information on the motion.

’675 Patent at 20:63-21:8 (emphasis added); *see also id.* at 41:52-65. In other words, the claimed first period of time must be “selected (or specified)” and must be associated with a “time when the electronic device **is worn** on (or close to) a part of the user’s body.”⁵ Yet, the specification fails to

⁵ The claim limitation preceding the language of claim 13 at issue here also suggests that the first period encompasses a period when a user is wearing the device. That language describes the first step of the method as “determining whether an electronic device is worn.” ’675 Patent at 46:32-34. The parenthetical phrase at the conclusion of the language in dispute could also support a

provide any guidance as to how to select or specify the first period of time, thereby allowing, for example, Complainant to subjectively select any period of time to satisfy the claimed “first period of time” requirement. Rosing Dec. at ¶¶ 45, 48; *see Interval Licensing*, 766 F.3d at 1373 (“Where, as here, we are faced with a ‘purely subjective’ claim phrase, we must look to the written description for guidance.”).

Similar issues arise when attempting to define the term “second period of time.” The specification suggests that this period of time also references a selected or specified period of time when it explains that “the second period of time may be shorter than the first period of time.” ’675 Patent at 21:30-31. This is a comparison a skilled artisan cannot undertake if the language of the claim allows defining time periods by the status of the electronic device relative to the user, *e.g.*, either worn or not worn, or the state of the user, *e.g.*, awake or asleep. Rosing Dec. at ¶ 46.

2. The Terms of Degree—“Shortly Before and/or Immediately After” Are Indefinite

While the deficiencies with the “first time period” and “second time period” terms are sufficient to hold the claim term indefinite, they do not end the problems with the claim. Further ambiguity arises with the need to understand a temporal relationship between these undefined time periods. The “second period of time” is required to occur “shortly before and/or immediately after [the] first period of time.” Courts have consistently held that phrases like “shortly before and/or immediately after” constitute terms of degree which, while not inherently indefinite must provide objective boundaries for persons of ordinary skill in the art when read in the context of the patent’s specification and prosecution history. *See, e.g., Intell. Ventures I LLC v. Symantec Corp.*, No. 13-

conclusion that the first period of time occurs “when it is determined that the electronic device is worn on a part of the user’s body.” *Id.* at 46:38-40. The final parenthetical phrase adds further confusion as to whether both the first and second period of time occur when a user is wearing the ring. Rosing Dec. at ¶ 47.

cv-440-LPS, 2016 WL 948879, at *4 (D. Del. Mar. 10, 2016) (holding that while the temporal term “substantial concurrence refers to length of delay,” the term is not indefinite because “whether the length of delay meets the limitation can be ascertained by reference to the cause of delay,” as disclosed in the intrinsic evidence).

When a word of degree is used in a patent claim, “the court must determine whether the patent provides some standard for measuring that degree.” *Biosig Instruments, Inc. v. Nautilus, Inc.*, 783 F.3d 1374, 1378 (Fed. Cir. 2015). The Federal Circuit requires that terms of degree, “when read in light of the specification and the prosecution history,” provide objective boundaries for those of skill in the art. *Liberty Ammunition, Inc. v. United States*, 835 F.3d 1388, 1396 (Fed. Cir. 2016). The written description is “key to determining whether a term of degree is indefinite.” *Akamai Tech., Inc. v. MediaPointe, Inc.*, 159 F.4th 1370, 1378 (Fed. Cir. 2025). A term of degree fails the definiteness requirement if it “depend[] ‘on the unpredictable vagaries of any one person’s opinion’” or is “purely subjective.” *Niazi Licensing Corp.*, 30 F.4th at 1348.

Reviewing the specification with an eye towards understanding objective boundaries for the terms “shortly before” or “immediately after” provides no insight into an objective temporal relationship between the occurrence of the “first period of time” and the “second period of time.” Specifically, the specification and prosecution history fail to provide objective boundaries for a POSITA to determine the “first period of time,” which in turn renders the phrase “shortly before and/or immediately after” an indefinite term of degree. For example, if the first period of time encompasses several days—as could be the case in the way that Complainant has alleged, *e.g.*, any period of time when the electronic ring is not being worn—then a person of ordinary skill in the art may consider an event that occurred a few hours before or after the first period of time ended as meeting the “shortly before” or “immediately after” requirement. But should the length of the

first period of time change, *e.g.*, from a few days to a few minutes, then it seems “shortly before” or “immediately after” would be measured in seconds and not hours. With no clear boundaries defining the first period of time, determining whether an event occurs “shortly before and/or immediately after” is merely subjective. *Rosing Dec.* at ¶ 49.

The intrinsic record is not devoid of any references to time let alone any specificity regarding exactly when something would occur according to the claimed terms of degree. The ’675 Patent includes a discussion of an alternative that sets periods over which the device may seek to take certain biometric signals:

According to various embodiments, the measurement module 506 may perform a measurement operation when a biometric information measurement period (e.g., about 1 minute) arrives. According to various embodiments, the measurement module 506 may perform the measurement operation based on the motion of the electronic device or a user when the biometric information measurement period arrives. According to an embodiment, when the measurement period arrives, the measurement module 506 may identify the motion of the user or the electronic device that was measured before the period arrives, and when the motion satisfies a selected (or specified) condition (e.g., when the strength of the motion is lower than a selected (or specified) strength), the measurement module 506 may measure biometric information. According to another embodiment, when the motion of the electronic device or the user does not satisfy the selected (or specified) condition (e.g., when the strength of the motion is higher than the selected (or specified) strength), **the measurement module 506 may monitor a change in the motion for a predetermined period of time (e.g., about 10 seconds) with respect to the time when the period arrives.** According to an embodiment, when the change in the motion for the selected (or specified) period of time satisfies the selected (or specified) condition (e.g., when the average of motion data is smaller than a selected (or specified) value), the measurement module 506 may measure biometric information.

’675 Patent at 18:19-45 (emphasis added). This text suggests defining a set period of time (*e.g.*, 1 minute) to take a biometric measurement period and a set period of time to take another measurement. *See, e.g.*, Figure 5B (time periods defined as 520 associated with a first timer and 524 associated with a second timer), 19:35-20:6. However, it is unclear how this language relates to the claim language because the claim language discusses obtaining information about the motion

of the device while the specification describes the first time period as a biological measurement period—something that requires the use of the second sensor. Rosing Dec. at ¶ 50. But the claim requires the “activating” the second sensor based upon information acquired shortly before or immediately after the first period. It is unclear how the biological measurement period (520) can occur while the second sensor is inactive. Rosing Dec. at ¶ 50.

In the next section, Respondents offer the argument that claim 15 fails to meet the definiteness requirements. As courts have held that language of dependent claims is relevant to definiteness analysis, Respondents show how claim 15 supports an argument that this tribunal should hold claim 13 invalid under 35 U.S.C. § 112(b). *See, e.g., Enzo Biochem, Inc. v. Applera Corp.*, 599 F.3d 1325, 1334-1335 (Fed. Cir. 2010). Claim 15 requires “acquiring additional information on the motion of the electronic device using at least a part of the at least one first sensor for **a third period of time, which is shorter than the first period of time**, immediately after the second period of time passes.” ’675 Patent at 46:58-62 (emphasis added). Again, in this instance, the claim requires a bounded first period of time to conclude that a third period of time is shorter than the first period of time. Yet, nothing in the intrinsic evidence suggests how the required first period of time is determined let alone what is meant by “shorter.” The specification suggests that the first period of time, second period of time, and third period of time are selected or specified time ranges, but neither the claim language nor the specification provides the reasonable certainty to define those times. *See, e.g.,* ’675 patent at 41:51-67, 42:8-23, 42:30-37, 42:57-68; Rosing Dec. at ¶ 57.

As noted at the outset, Complainant’s proposed construction fails to cure the deficiencies discussed above and is fatally flawed in that it ignores the canons of claim construction in favor of writing out the “shortly” portion of the “shortly before” phrase. Complainant proposes that the

phase “second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user’s body” mean “second period of time which begins: (i) **before** the first period of time has ended when it is determined that the electronic device is worn, or (ii) immediately after the first period of time has ended when it is determined that the electronic device is worn.” (emphasis added) This construction, however, does nothing to address the ambiguities associated with the “first period of time” and “second period of time” phrases or define the terms of degree. Moreover, Complainant’s proposed construction improperly removes one of the terms of degree (“shortly”) and broaden the scope of the claim to allow for the second period of time to occur during the first period of time, as opposed to being a separate and distinct time period, as required by the claim. *See Merck & Co., Inc. v. Teva Pharma. USA, Inc.*, 395 F.3d 1364, 1372 (Fed. Cir. 2005) (“A claim construction that gives meaning to all the terms of the claim is preferred over one that does not do so.”); *General Am. Transp. Corp. v. Cryo-Trans, Inc.*, 93 F.3d 766, 770 (Fed. Cir. 1996) (rejecting a claim construction that was inconsistent with the specification and rendered claim requirements superfluous).

In short, the term “acquiring first information on a motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user's body” is indefinite twice over.

B. Term No. 2: “acquiring additional information on the motion of the electronic device using at least a part of the at least one first sensor for a third period of time, which is shorter than the first period of time, immediately after the second period of time passes; and activating the at least one second sensor based on at least a part of the additional information”

Claim Term	Complainant’s Construction	Respondents’ Construction
“acquiring additional information on the motion of the electronic device using at least a part of the at least one first sensor for a third period of time, which is shorter than the first period of time, immediately after the second period of time passes; and activating the at least one second sensor based on at least a part of the additional information” Claim 15 of the ’675 Patent	Within this phrase, “third period of time” means: “period of time that immediately follows the second period of time and during which additional information regarding the motion of the electronic device is collected by at least a part of the at least one first sensor and used to determine whether to activate the at least one second sensor”	Indefinite

The intrinsic evidence fails to suggest the meaning of the term “acquiring additional information on the motion of the electronic device using at least a part of the at least one first sensor for a third period of time, which is shorter than the first period of time, immediately after the second period of time passes; and activating the at least one second sensor based on at least a part of the additional information,” rendering claim 15 indefinite. “The claims, when read in light of the specification and the prosecution history, must provide objective boundaries for those of skill in the art.” *Interval Licensing*, 766 F.3d at 1371 (citing *Nautilus, Inc.*, 572 U.S. at 911). As explained above with respect to Disputed Term No. 1, the specification and the prosecution history fail to provide objective boundaries for the claimed “first period of time” and therefore fail to provide the necessary disclosure for determining the required temporal relationship between the “first period of time” and the “third period of time.”

To begin, Complainant asserts that:

The Oura Accused Product, during a third period of time beginning immediately after the second period of time (e.g., the period of time ending after the user wakes up), which may be shorter than a first period of time when the Galaxy Watch (sic) product is not being worn, acquires additional information on the motion of the electronic device using the at least one first sensor, e.g., an accelerometer.

Ex. 2 at 36-37. Here, Complainant alleges that the third period of time is *any* period after which a user wakes up based upon the assertion that the second period of time ends when a user wakes up. This allegation is purely subjective and unsupported by the specification. But more than that, it demonstrates that the claimed third period of time is subject to the “unpredictable vagaries of any one person's opinion.” *See Niazi Licensing Corp.*, 30 F.4th at 1348.

For example, the '675 Patent explains:

According to various embodiments, the instructions may be set to configure the processor to: acquire additional information on the motion of the electronic device using at least a part of the at least one first sensor for a third period of time, which is shorter than the first period of time, immediately after the second period of time passes.

'675 Patent, 21:41-46; *see also* '675 Patent, 42:7-12.

According to various embodiments, at least one of the second period of time and the third period of time may be set to include the time point when the first period of time passes.

'675 Patent, 21:51-54, *see also* '675 Patent, 42:18-20. Here, the only objective boundaries related to the third period of time are that it must be shorter than the first period of time. However, without any objective boundaries to define the first period of time, the third period of time is also indefinite and subject to the same shortcomings of the first period of time. In short, this term is indefinite.

C. Term No. 3: “biometric signal”

Claim Term	Complainant’s Construction	Respondents’ Construction
“biometric signal” Claim 14 of the ’359 Patent	Correcting “the biometric signal” to be “the biological signal” and applying the plain and ordinary meaning (<i>i.e.</i> , a signal that is associated with biological information of the user).	Correcting “the biometric signal” to be “a biometric signal” and applying the plain and ordinary meaning (<i>i.e.</i> , a signal relating to the identification of an individual based on physical characteristics, such as fingerprints, finger lengths, facial features, retinal patterns, voice recognition, and weight).

The constructions proposed by both parties seek to cure the error that exists in the ’359 Patent—the lack of antecedent basis for the term “the biometric signal.” ’359 Patent at 15:13-26. However, Samsung’s proposed construction seeks to do more than correct the antecedent basis: it seeks to change the *term* and its *meaning*. Courts are permitted to “correct ‘obvious minor typographical and clerical errors in patents,’” but “only if (1) the correction is not subject to reasonable debate based on consideration of the claim language and the specification and (2) the prosecution history does not suggest a different interpretation of the claims.” *Pavo Solutions LLC v. Kingston Tech. Co., Inc.*, 35 F.4th 1367, 1373 (Fed. Cir. 2022) (citations omitted). The error must also “be ‘evident from the face of the patent,’ and the determination ‘must be made from the point of view of one skilled in the art.’” *Id.* (citations omitted). Further, when “deciding whether a particular correction is appropriate, the court ‘must consider how a potential correction would impact the scope of a claim and if the inventor is entitled to the resulting claim scope based on the written description of the patent.’” *Id.* (citations omitted).

Samsung acknowledges the lack of antecedent basis in the claim, but in its proposed construction it seeks to also *amend* the claim language and its interpretation when there is no basis

for doing so. Samsung’s proposed construction should be rejected. *See Chef Am., Inc.*, 358 F.3d at 1374. There is only one reasonable correction of the claim (if the Court is inclined to save the claim by doing so): substituting “**the**” for “**a**” before “biometric signal.” *See* ’359 Patent at 15:14-27 (emphasis added). This correction not only fixes the clear error on “the face of the patent,” *Pavo Solutions*, 35 F.4th at 1272, but also preserves the claim as drafted by the patentee. *Chef Am., Inc.*, 358 F.3d at 1347 (explaining that the canons of claim construction require courts to “construe claims based on the patentee’s version of the claim as he himself drafted it” and that it is “settled practice” to “construe the claim as written, not as the patentees wish they had written it.”). That Samsung understood it was claiming a “biometric signal” is clear from the intrinsic record. For example, in response to the Non-Final Office Action mailed June 21, 2019, Samsung amended claims but notably left un-amended the claimed “biometric signal”:

15. (Currently amended) A wearable device comprising:
a motion sensor;
a biological sensor; and
a controller configured to:
~~obtain, with respect to an object external to the wearable device, a movement signal of the~~
~~an object external to the wearable device using the motion sensor; and~~
obtain a biological signal of the object using the biological sensor;
determine a select period of the movement signal that meets a specified condition;
determine, as a valid signal, at least a portion of the biometric signal that is determined to
~~correspond to the determined corresponding to a select period that the movement is smaller than a~~
~~specified condition as a valid signal; and~~
perform a specified function using the valid signal.

Appendix A to Complaint at 156.

The phrase “biometric signal,” as understood by a POSITA, has a plain and ordinary meaning separate and distinct from the term “biological signal.” As explained by Dr. Rosing, a POSITA would understand this distinction. Rosing Dec. at ¶ 60. Specifically, a POSITA would understand that a biological signal is a signal that relates to “muscle contraction or muscle relaxation of a body part, such as a wrist or a heart,” ’359 Patent at 3:56-58, and that a biometric

signal is a signal that relates to the identification of an individual based on physical characteristics, such as fingerprints, finger lengths, facial features, retinal patterns, voice recognition, and weight.⁶

Rosing Dec. at ¶ 60.

Samsung claimed the term “biometric” in claim 14, and its plain meaning should apply. *See W.E. Hall Co., Inc. v. Atlanta Corrugating, LLC*, 370 F.3d 1343, 1350 (Fed. Cir. 2004) (explaining that the plain and ordinary meaning should be applied if “nothing in the specification or the prosecution history clearly suggests that [the inventor] intended to use the terms in a manner other than according to their ordinary meanings.”).

In short, the term “the biometric signal” should be corrected to be “a biometric signal” and construed according to its plain and ordinary meaning. Any attempt to change the term to “the biological signal” is nothing more than an improper attempt to broaden the scope and meaning of the claim.

⁶ *See* The American Heritage Dictionary of the English Language, 5th ed. (2011), p. 184 (Ex. 3) (defining biometrics as “[t]he measurement of physical characteristics, such as fingerprints, DNA, or retinal patterns, for use in verifying the identity of individuals.”); Dictionary of Computer Vision and Image Processing (2005), p. 32 (Ex. 4) (defining biometrics as “[t]he science of discriminating individuals from accurate measurement of their physical features. Example biometric measurements are retinal lines, finger lengths, fingerprints, voice characteristics and facial features.”); Dictionary of Information Security (2006), p. 24 (Ex. 5) (defining biometric at “pertaining to the measurement of the human body: in security terms, relating to means of authentication based on patterns unique to an individual’s body, such as fingerprints and retinal patterns, or behaviour, such as voiceprints or handwriting.”); Merriam-Webster’s Collegiate Dictionary, 11th ed. (2012), p. 124 (Ex. 6) (defining biometric as “of, relating to, or utilizing biometrics or biometry” and defining biometrics as “the measurement and analysis of unique physical or behavioral characteristics (as fingerprint or voice patterns) esp. as a means of verifying personal identity.”); Newton’s Telecom Dictionary, 27th ed. (2013), p. 203-204 (Ex. 7) (defining biometrics as “the study of unique and measurable physical (or biological) characteristics such as fingerprints, retinal patterns, or handwriting . . .” and defining biometric device as a device that “authenticates a user by measuring some hard-to-forge physical characteristic, such as a fingerprint or retinal scan.”).

VI. CONCLUSION

For the reasons above, the ALJ should adopt Oura's proposed constructions.

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Respectfully submitted,

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